

High Impact Fantasy Football Decisions

Motivation

Fantasy Football is an exciting game that is a perfect sandbox for analytics and data exploration. The game is based around leagues. The leagues are made up of teams. Each team has a manager who makes roster decisions with the goal of maximizing their team's total points. The points are generated based on the real performance of NFL players during a weekly football game. There are four decision categories that a team manager must consider: draft decisions, waiver wire decisions, start/sit decisions, and trade decisions. This analysis wants to explore which decisions have the most impact. What trends are associated with these decisions? Can we use additional data outside of the Fantasy Football game to improve decision making?

To compare these different decision criteria, we need to analyze them using like-terms. One way to do that is to view each decision through the lens of "points added". If a player is drafted to a team, that player added some amount of points to that team's total score. Likewise, if a player is added or dropped from a team via waiver wire, that decision affected the team's total points at the end of the season. Each of the decision categories can be thought of with regards to "points added", and that means that we can effectively compare them.

There is an additional complication to this analysis. Managers are given many tools to improve their quality of play. There is an autodraft feature, there are weekly point projections for each player, and there are even trade analyzers built directly into the app. With this level of support, it becomes quite easy to be a competent Fantasy Football manager. So, we want to analyze how much managers are able to set themselves apart from this minimally competent baseline. To do that, we defined a baseline for each of the four decision criteria, and we will evaluate data from past leagues to see how they performed relative to that baseline.

The primary research questions are:

1. Which decision criteria is the most impactful?
2. What trends are associated with each decision type?
3. Can we use additional data outside of the Fantasy Football application to improve our decision making?

Data Sources

ESPN Fantasy Football API Data

We collected league data using the public ESPN Fantasy Football API accessed through the `espn_api` Python package. The first step is to find league ID's for public leagues, that match our desired format. That was quite difficult, because ESPN doesn't display that information. So, we selected a likely range for ESPN league ID's and tried random League ID's until we got enough leagues to make a reasonable sample size. Data was retrieved for approximately 500 leagues across the 2021 start year– 2025 end year seasons, resulting in 1,467 unique league-years.

The following datasets were extracted:

Draft data including league ID, season, player name, drafting team, round, pick number, and autodraft indicator. Weekly roster snapshots were used to infer much of the following data: Lineup data including week, team, player, roster slot, weekly points, and starter designation. Transaction data including week, team, player, and action type (add, drop, trade). All three of those datasets were stored as a CSV and organized by year.

We pulled injury information via a python package that has an object method to import data as a pandas dataframe. Injury data contains data like player last name, game status, injury description, etc. That was also stored as a CSV.

Additionally, we investigated play-by-play data from NFLSavant, but found that it was likely out of scope for this project. It contained 10-12 MB of statistics like game data, minute, second, offensive team, down to go, yard line, description, etc..

Derived Baseline Datasets

Three baseline datasets were constructed from the lineup and draft data.

The autodraft baseline was created by filtering leagues in which all teams were marked as autodrafted to estimate expected season-long production by draft pick. Points were only counted towards the team if they were part of the optimal line-up for any week. This requires the team formulation to be legitimately playable. Otherwise, the analysis would show that all-QB leagues were best. Additionally, there were two methods of regression used to smooth or denoise the baseline. These smoothed trend-lines were what we used as the baseline.

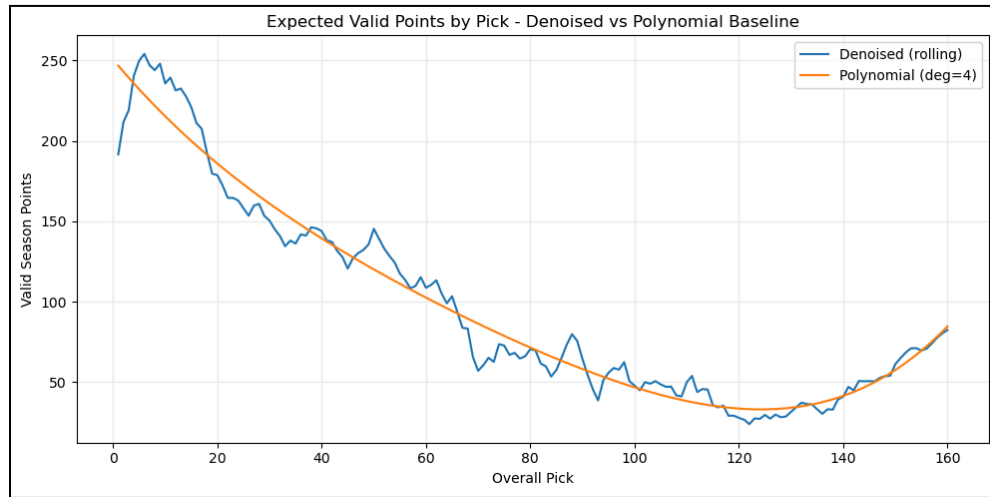


Figure 1: Autodraft Baseline- Valid Points Added by Pick for all Seasons

Surprisingly, the first several autodrafted picks were underperforming. It was a serious trend that spanned several seasons. We investigated why, and thought that there may be real “football” related reasons (For example, high usage rate players may get injured more). However, it may all be explained by the autodraft variance. When it comes to autodraft picks, there is less variance at the top of the draft, and therefore, they are more susceptible to dramatic injury-based underperformance.

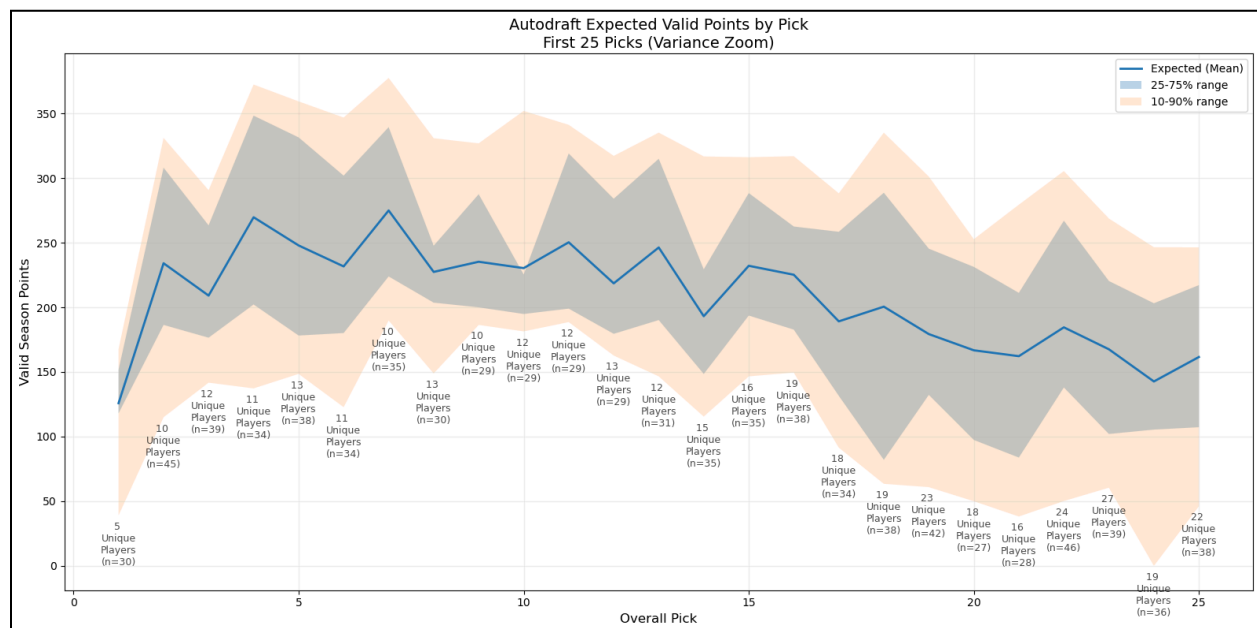


Figure 2: Autodraft Baseline- First 25 Picks, Variance, and Number of Unique Players All Seasons

The waiver wire baseline was defined as the bottom 25th percentile, valid, weekly points per contributing player-week. Basically, “What does a bottom-25% manager typically produce on waiver wire, per active

waiver stint-week?”. This was a difficult baseline to define. There is no “autodraft equivalent” for waiver wire. We tried to select a baseline that assumes a minimally competent manager. We tried other baselines, but the bottom 25th percentile felt most appropriately comparable to a full autodraft in terms of manager skill.

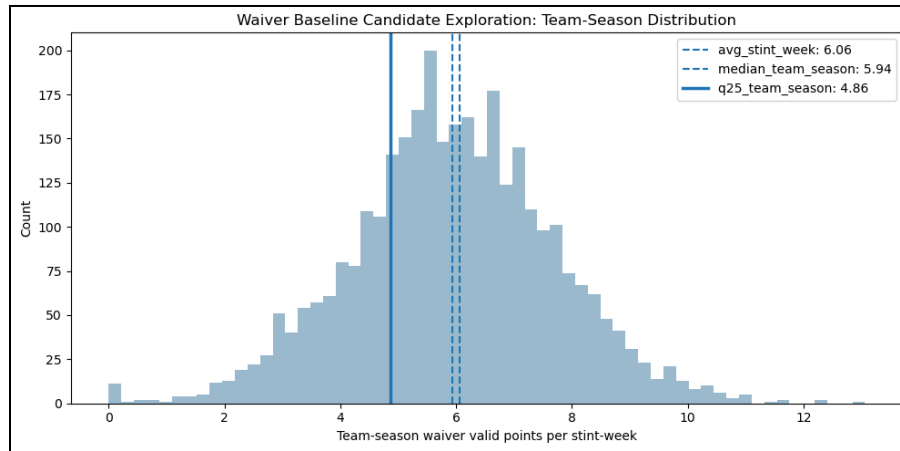


Figure 3: Q25 Waiver Wire Baseline

The Start/Sit baseline was very easy to define. ESPN makes projections about which players will have the most points on any given team each week. In line with our “minimally competent manager theme”, we decided to set the baseline to equal the optimum projected lineups.

For Trades, we decided that the baseline should be no trade points added at all. Trades are actually quite rare in fantasy football, making up less than 3% of all Fantasy Football actions.

For injuries, the baseline was derived by getting the players past historical performance against their past projected points. The effect of injury can then be calculated based on how far off their injured performances are from that expectation.

Data Manipulation Methods

Data Cleaning

Player names were standardized across datasets to enable reliable joins. Duplicate records were removed, and missing weekly scores were imputed as zero when a player was inactive. Positional slot labels were normalized to enforce consistent roster constraints.

Additionally, all of the different league formats were filtered to match our “standard” league definition. This required a variety of simple and creative filters to ensure PPR (points per reception), league size, draft length, positional requirements, positional flexibility, and additional scoring rules. Our final standardized leagues should only include:

- QB: 1, RB: 2, WR: 2, TE: 1, FLEX: 1, K: 1, D/ST: 1, OP: 0
- Total starters: 9
- Draft length: 160 picks
- 1 Point-PPR

Optimal Lineup Reconstruction

To remove start/sit noise, an optimal lineup was reconstructed for each team-week by selecting the highest-scoring eligible players subject to positional limits. A binary indicator, `SelectedOptimal`, was assigned, and valid weekly points were defined as points scored when `SelectedOptimal` was true.

Draft Attribution

Drafted players were linked to their season-long production and assigned to their original drafting team. Transaction logs were used to identify trades and prevent misclassification of player movement.

Waiver Attribution

Players not originally drafted by a team and added via waiver or free agency actions were classified as waiver acquisitions. These records were joined with lineup data to compute valid weekly contributions from waiver players.

Dataset Integration

All datasets were joined using league ID, team, player, and week to produce a final panel dataset containing all of our advanced metrics.

Dataset Cleaning for Injury Analysis

When the injury data is brought in, the injury type has to be normalized. The status shows left hand, right hand, and hand. All of these injury types are consolidated down to just hand. After normalizing the injury types, the player names must be normalized. This is done by stripping out excess space characters, removing suffixes such as Jr and Sr. Finally, while doing calculations the two datasets used have different naming nomenclature for position. This is handled by creating another column in the dataframe named like the other dataframe prior to merging with the calculated player baselines.

Analysis

Draft Performance Relative to Autodraft

Total valid draft points above expected were calculated for each team and examined by draft round and total. On average, humans were outperforming autodraft by a measurable amount with both the polynomial regression baseline and the denoised baseline.

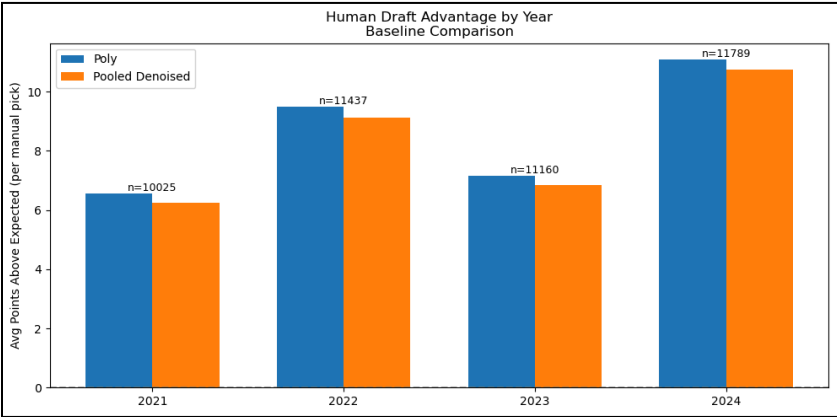


Figure 4: Humans Outperform Autodraft

Autodraft generated an average of 1651.9 valid points and humans generated an average of 1674.8 valid points per season.

Waiver Wire Contributions

Waiver value had some very interesting trends. Firstly, week 15 had a consistent trend of underperformance. In many leagues, week 15 is the first week of playoffs. This dip could be caused by desperate team managers making bad decisions; troublemakers who drop their good players when they miss playoffs as a joke; or potentially bye-weeks breaking our algorithms. For the purposes of expediting our investigation, we will drop week 15 from the analysis, but it deserves future investigation. Figure 5 displays the points added over baseline expectation in each week (bars) via draft and waiver wire. Additionally, it shows the cumulative effect of each waiver addition with a line graph and secondary y-axis.

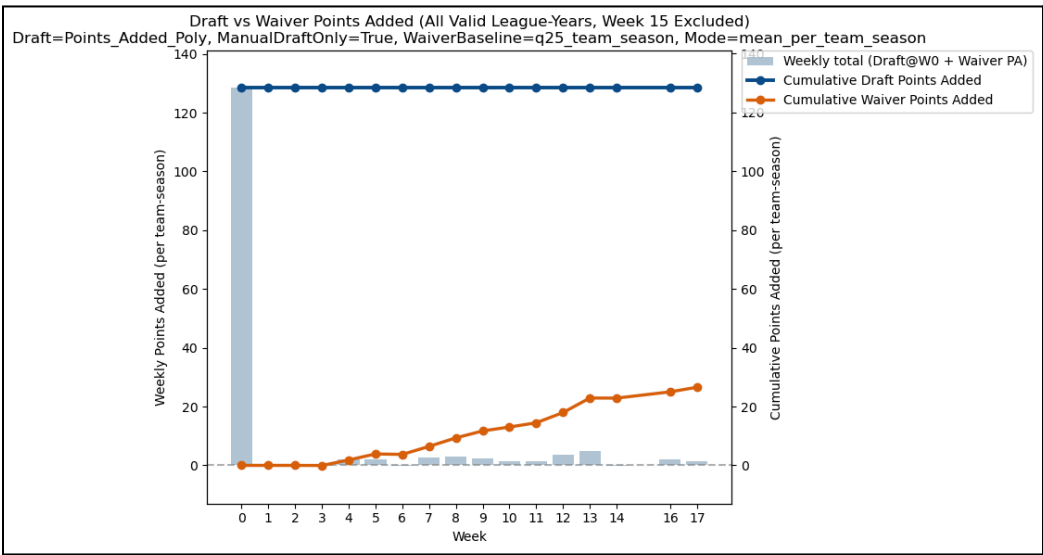


Figure 6: Draft Vs Waiver Wire- Points Over Baseline Expectation All seasons, Ignore Week 15

The data is quite clear that waiver wire decisions do not come close to adding as many points as draft decisions. However, waiver wire additions are fascinating, because they are the primary tool for gaining, losing, or recovering points during the fantasy season. Figure 7 shows how waiver wire decisions shaped the outcome of one league.

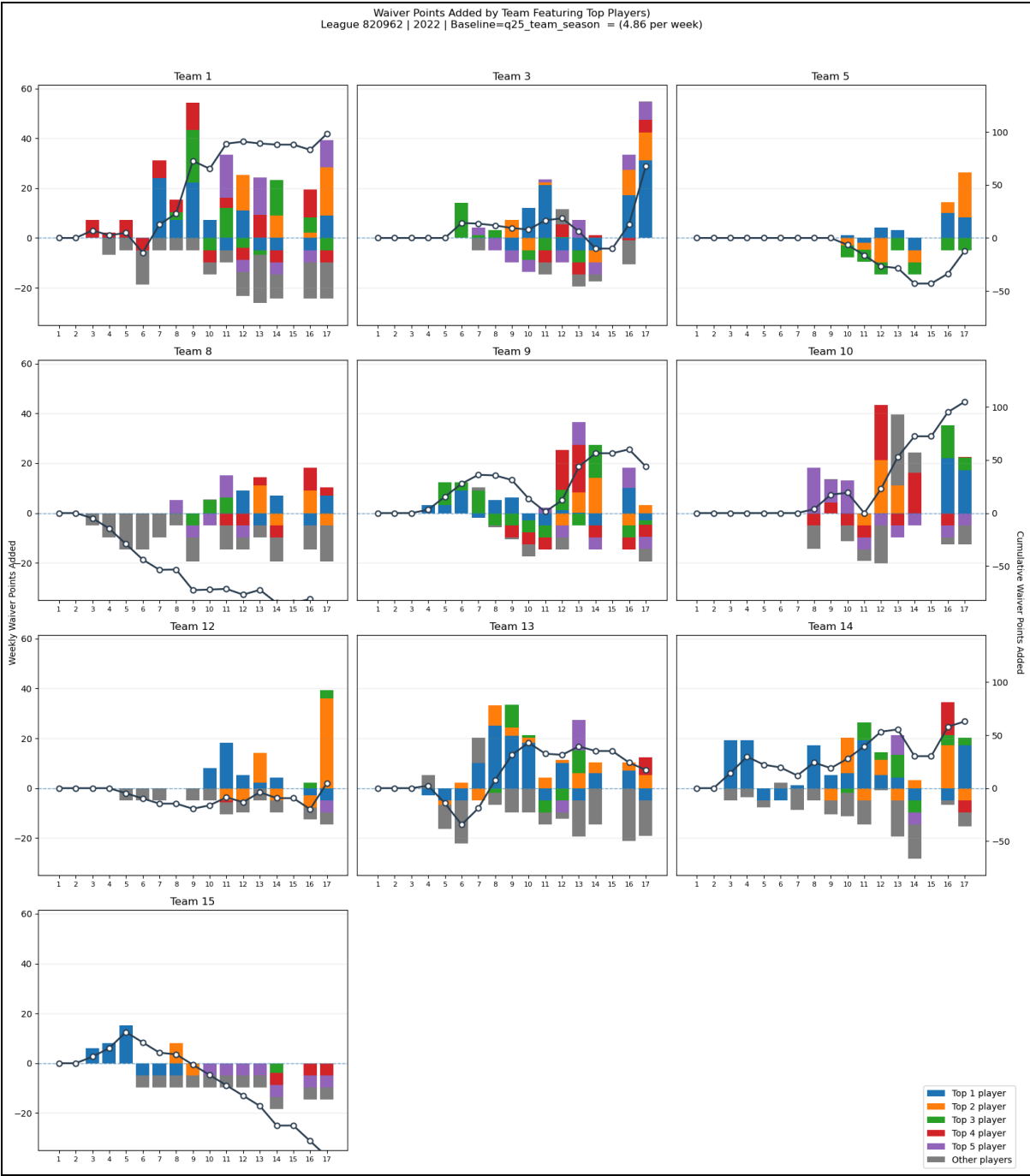


Figure 7: Points Added via Waiver Wire- League/Top Player View

Waiver wire decisions do not account for the most total points added to a team during a season, but they drive the fun, engaging, dynamism that we look for in Fantasy Football. Furthermore, waiver wire decisions can be more impactful when you consider them from a “per decision” standpoint. A draft night is full of decisions, typically more decisions are made on draft night than are made on waiver wire over the course of a season. Figure 8 shows how draft and waiver decisions stack up when normalized for decision count. From that standpoint, waiver wire decisions may give managers the most value for their research.

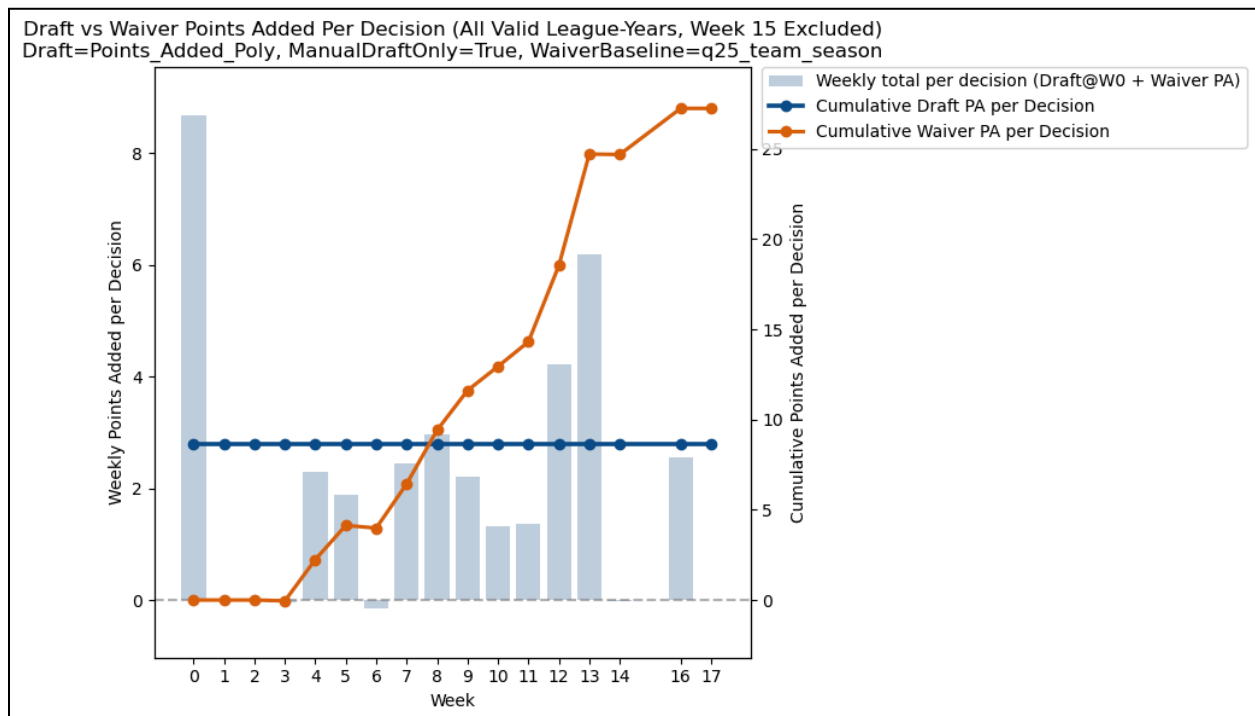


Figure 8: Draft Vs Waiver Wire- Points Over Baseline per Decision, All Seasons, Ignore Wk 15

Manual Start/Sit Decisions vs Optimal Projected Point Lineups

Good draft and waiver wire decisions get good players on the team, but none of that matters if the right players aren't starting. To this point, we've assumed that the manager played the optimal lineup. To analyze start/sit lineups, we will compare the actual lineup's performance to the optimal projected lineup's performance.

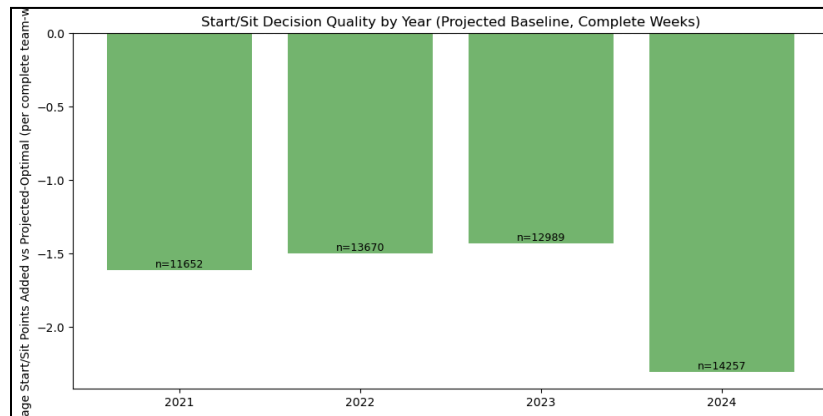


Figure 8: Manual Start/Sit vs Optimal Projected

Figure 8 shows a pretty convincing trend. In every year, on average, players lost points when they deviated from the projected optimal starting lineup.

Analysis of Trades

The truth about trades is that they are so rare fantasy managers shouldn't rely on them. Making up less than 3% of all Fantasy Football actions, it's safe to say that they're the least impactful decision category of the four.

Effects of Injury on Actual vs Projected Points

Three different comparisons were made to examine how injuries impacted expected points. Position x Status, Injury Type x Status, and Injury Type x Position were looked at. The analysis showed that players designated as Questionable have a high variability in point differential against projected while doubtful and out had little variability. Some injuries also had highly unfavorable outcomes such as rib injuries for wide receivers. Unexpectedly some injuries had positive outcomes on expected point totals such as a foot injury for tight ends. The key takeaway here is that starting players with a questionable status comes with high variability in expected points while others should be more consistent.

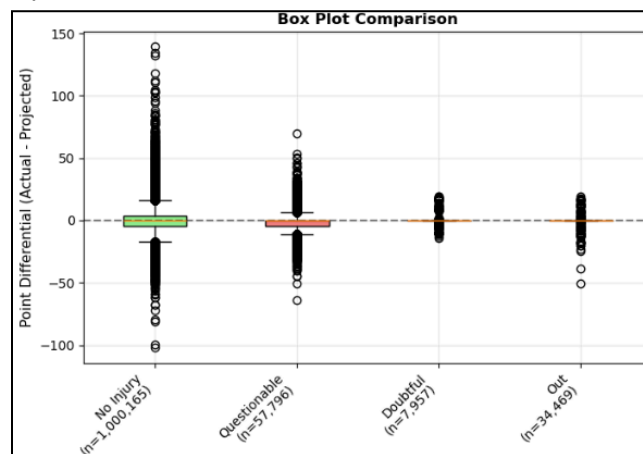


Figure 9: Injury Status and Variation from Projection

Conclusion

This project demonstrates that a “points added above baseline” framework can be used to compare a variety of decisions in Fantasy Football. The analysis is pretty clear that the draft is the decision category that can earn the most points for your team, but waiver wire can have the biggest impact per decision. Furthermore, the projected lineups are well-optimized and trades are too rare to be regularly impactful. This is clearly represented in Figure 10:

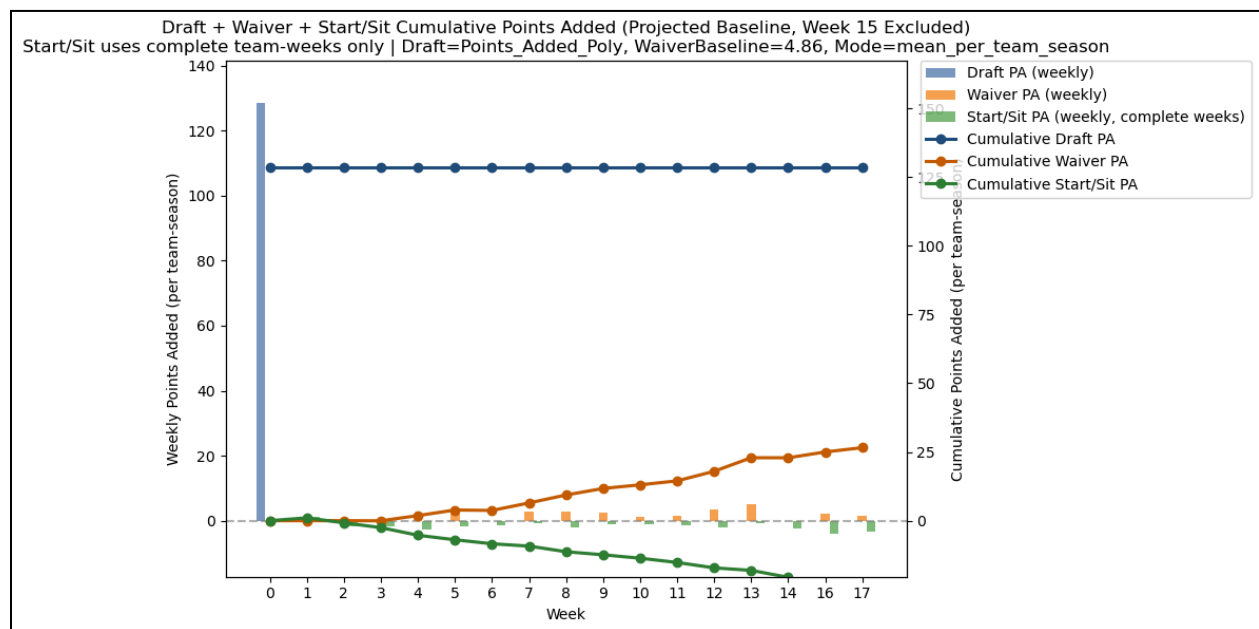


Figure 10: Cumulative Points Added over Baseline, All Seasons, All Decision Categories

Additionally, our work on injury analysis demonstrated that further analysis may be capable of improving fantasy football decisions, but it can be a challenge.

Statement of Work

Tristan Morgan

Describe contributions:

Fantasy Football API exploration, general requests, and transaction inference from roster snapshots.

Acquired Injury database and analyzed it.

General debugging and collaboration throughout.

Refactored and reorganized code to fit in Notebook.

David Brand

Describe contributions:

Fantasy Football API exploration, general requests, and League ID search.

Pulled NFL Savant Play-by-Play Data. Unused.

Defined baselines and points added visualizations.

Collaboration:

We had an effective collaboration process. We communicated regularly and openly about our progress and next steps. We started initial exploration in a Google Colab environment, but quickly switched to GitHub. Initially, we struggled to parallelize our work, but once we got all of the ESPN data pulled, we were able to work separate tasks at the same time.

Future Improvements:

We could have been more organized regarding the parallelization of tasks and the delivery dates of milestones. We agreed that we should have used a Gantt chart to organize the process in the beginning. Furthermore, we underestimated the difficulty of using the unofficial, undocumented ESPN API.
